12.3 #6 Let f be continuous over $\mathbb{R}$ s.t.

$$F(k)=0 \text{ for } |k|>\pi.$$

(a) Show that $f(x)=\sum_{n=-\infty}^{\infty} f(n)\cdot\frac{\sin[\pi(x-n)]}{\pi(x-n)}$. (*)

By Fourier inversion, (cf Strass, Page 344 (31)).

$$f(x)=\int_{-\infty}^{\infty} F(k)e^{ikx}\cdot\frac{dk}{2\pi}. \quad (1)$$

(actually we are assuming $f\in L^1(\mathbb{R})$).

Note that the nth Fourier coefficient of $F(k)$ on $[-\pi,\pi]$ is

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} F(k)e^{-in\pi k/\pi}dk=f(-n), \text{ by } (1).$$

Thus $F(k)=\sum_{n=-\infty}^{\infty}\hat{F}(n)\cdot e^{ink}=\sum_{n=-\infty}^{\infty} f(-n)\cdot e^{ink}=\sum_{n=-\infty}^{\infty} f(n)e^{-ink}$ (2)

Plugging (2) into (1), and note $F(k)=0$ for $|k|>\pi$, then

$$f(x)=\int_{-\pi}^{\pi}\left(\sum_{n=-\infty}^{\infty} f(n)\cdot e^{-ink}\right)e^{ikx}\cdot\frac{dk}{2\pi}$$

by Fubini

$$=\sum_{n=-\infty}^{\infty}\left(\int_{-\pi}^{\pi} f(n)\cdot e^{-ink}\cdot e^{ikx}\cdot\frac{dk}{2\pi}\right)$$

$$=\sum_{n=-\infty}^{\infty}\left[f(n)\cdot\left(\int_{-\pi}^{\pi} e^{i(x-n)k}\cdot\frac{dk}{2\pi}\right)\right]$$

$$=\sum_{n=-\infty}^{\infty}\left[f(n)\cdot\frac{1}{i(x-n)}e^{i(x-n)\cdot k}\cdot\frac{1}{2\pi}\Big|_{k=-\pi}^{k=+\pi}\right]$$

$$= \sum_{n=-\infty}^{\infty} f(n) \cdot \frac{\sin[\pi(x-n)]}{\pi(x-n)}$$

(If you are bit careful. the above formula only holds for $x \neq 0$. If $x=0$, then the right hand is simply $f(0)$.)

(b) Let $F(k)=1$ over $(-\pi,\pi)$ and $F(k)=0$ if $|k|>\pi$, then by (1),

$$f(x) = \int_{-\infty}^{\infty} F(k) e^{ikx} \cdot \frac{dk}{2\pi} = \int_{-\pi}^{\pi} \frac{e^{ikx}}{2\pi} dk$$

$$= \begin{cases} 1 & \text{if } x=0 \\ \frac{\sin(\pi x)}{\pi x} & \text{if } x \neq 0 \end{cases}$$

$$\Rightarrow f(n) = \begin{cases} 1 & \text{if } n=0 \\ 0 & \text{if } n \neq 0. \end{cases}$$

Thus if $x \neq 0$, LHS of $(*) = \frac{\sin(\pi x)}{\pi x}$

RHS of $(*) = \sum_{n=-\infty}^{\infty} f(n) \cdot \frac{\sin[\pi(x-n)]}{\pi(x-n)} = \frac{\sin[\pi(x-0)]}{\pi(x-0)} = \text{LHS}.$

$\boxtimes$

12.3 #7

(a) Let f be continuous over $\mathbb{R}$ and $\exists R>0$ s.t $f(x)=0$ for $|x|>R$.
Show that $g(x) := \sum_{n=-\infty}^{\infty} f(x+2\pi n)$ is periodic with period 2π.

Proof:

$$g(x+2\pi) = \sum_{n=-\infty}^{\infty} f(x+2\pi+2\pi n)$$

$$\overset{m=n+1}{=} \sum_{m=-\infty}^{\infty} f(x+2\pi m) = g(x)$$

(b) Show that the Fourier coefficients c_m of $g(x)$ on the interval $(-\pi,\pi)$ are $F(m)/2\pi$

Proof:

$$c_m = \frac{1}{2\pi}\int_{-\pi}^{\pi} g(k) e^{-im\pi k/\pi} dk$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} \sum_{n=-\infty}^{\infty} f(k+2\pi n)\cdot e^{-imk} dk$$

$$\overset{\text{Fubini}}{=} \frac{1}{2\pi} \sum_{n=-\infty}^{\infty}\left(\int_{-\pi}^{\pi} f(k+2\pi n)\cdot e^{-imk} dk\right)$$

$$\overset{y=k+2\pi n}{=} \frac{1}{2\pi} \sum_{n=-\infty}^{\infty}\left(\int_{-\pi+2\pi n}^{\pi+2\pi n} f(y)\cdot e^{im(y-2\pi n)} dy\right)$$

$$= \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} \left(\int_{-\pi+2\pi n}^{\pi+2\pi n} f(y) \cdot e^{-imy} \, dy \right)$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} f(y) e^{-imy} \, dy$$

$$= \frac{F(m)}{2\pi} \quad \text{by def'n of } F(m).$$

(c) The Fourier series of $g(x)$ on $(-\pi, \pi)$

is $g(x) = \sum_{m=-\infty}^{\infty} c_m \cdot e^{imx}$

By (a) and (b), we have

$$\sum_{n=-\infty}^{\infty} f(x+2\pi n) = \sum_{m=-\infty}^{\infty} \frac{F(m)}{2\pi} \cdot e^{imx}.$$

Letting $x=0$, then

$$\sum_{n=-\infty}^{\infty} f(2\pi n) = \sum_{m=-\infty}^{\infty} \frac{1}{2\pi} F(m) = \sum_{n=-\infty}^{\infty} \frac{1}{2\pi} F(n).$$

12.3 #9. Solve $-u_{xx}+a^2u=\delta$.

Denote the Fourier transform by $\widetilde{\mathcal{F}}$, then

$$-\widetilde{\mathcal{F}}(u_{xx})+a^2\widetilde{\mathcal{F}}(u)=\widetilde{\mathcal{F}}(\delta)$$

$$\Rightarrow -(ik)^2\widetilde{\mathcal{F}}(u)+a^2\widetilde{\mathcal{F}}(u)=1$$

$$\Rightarrow (k^2+a^2)\widetilde{\mathcal{F}}(u)=1$$

$$\Rightarrow \widetilde{\mathcal{F}}(u)=\frac{1}{k^2+a^2}.$$

By Fourier inversion,

$$u(x)=\widetilde{\mathcal{F}}^{-1}\left(\frac{1}{k^2+a^2}\right)=\frac{1}{2a}e^{-a|x|}$$

12.4 #2 Use Fourier transform to find u s.t.

$$\begin{cases} \Delta u = 0 & \text{in } \{(x,y)\in\mathbb{R}^2 \mid y>0\} \\ \dfrac{\partial u}{\partial y} = h(x) & \text{on } \{(x,y)\in\mathbb{R}^2 \mid y=0\} \end{cases}$$

Apply Fourier transform w.r.t. the x variable, then

$$\begin{cases} -k^2\,\hat{u}(k,y) + \dfrac{\partial^2}{\partial y^2}\hat{u}(k,y) = 0 \quad y>0 & (1) \\ \dfrac{\partial u}{\partial y}(x,0) = h(x) & (2) \end{cases}$$

By (1), for a fixed k,

$$\hat{u}(k,y) = A(k)e^{ky} + B(k)e^{-ky}$$

We may assume $\hat{u}(k,y) = C(k)\,e^{-|k|y}$ for some $C(k)$ to be determined.

On one hand,

$$\frac{\partial \hat{u}}{\partial y}(k,y) = \frac{\partial}{\partial y}\left(C(k)\cdot e^{-|k|\cdot y}\right) = C(k)\cdot -|k|\cdot e^{-|k|\cdot y}. \quad (3)$$

On the other hand, by Fourier transform to (2),

$$\frac{\partial \hat{u}}{\partial y}(k,0) = \hat{h}(k) \quad (4)$$

Letting $y=0$ in (3) and comparing with (4),

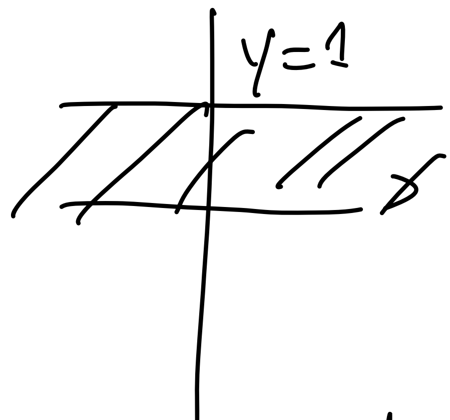
$$C(k) = -\,\hat{h}(k)/|k|.$$

Thus $\hat{u}(k,y) = -\dfrac{\hat{h}(k)}{|k|\cdot e^{|k|y}}$

$$u(x,y) = \mathcal{F}^{-1}(\hat{u}(k,y))$$
$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{u}(k,y)\, e^{ikx}\, dk.$$
$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} -\frac{\hat{h}(k)}{|k|\cdot e^{|k|y}} \cdot e^{ikx}\, dk.$$

12.4 #6.

$$\begin{cases} \Delta u = 0 & \{(x,y)\in\mathbb{R}^2 \mid 0<y<1\} \\ u(x,0) = 0 \\ u(x,1) = f(x) \end{cases}$$



Apply Fourier transform w.r.t. the x variable, then

$$\begin{cases} -k^2\, \hat{u}(k,y) + \frac{\partial^2}{\partial y^2}\hat{u}(k,y) = 0 \quad (1) \\ \hat{u}(k,0) = 0 \\ \hat{u}(k,1) = \hat{f}(k) \end{cases}$$

By (1), $\hat{u}(k,y) = A(k)e^{ky} + B(k)e^{-ky}$

By (2), $\hat{u}(k,0) = A(k)\cdot e^{k\cdot 0} + B(k)\cdot e^{-k\cdot 0} = 0$

$$\Rightarrow A(k) + B(k) = 0$$

Thus by (3), $\hat{u}(k,1) = A(k)e^{k\cdot 1} + B(k)\cdot e^{-k\cdot 1} = \hat{f}(k)$

$$\Rightarrow \quad A(k) = \hat{f}(k)/(e^{k} - e^{-k}) = \frac{\hat{f}(k)}{2\sinh(k)}.$$

Thus $\hat{u}(k,y) = \dfrac{\hat{f}(k)}{2\sinh(k)}(e^{ky} - e^{-ky})$

$$= \hat{f}(k)\frac{\sinh(ky)}{\sinh(k)}$$

$$u(x,y) = \mathcal{F}^{-1}(\hat{u}(k,y))$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{u}(k,y) e^{ikx}\, dk.$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{f}(k)\frac{\sinh(ky)}{\sinh(k)}\cdot e^{ikx}\, dk.$$